

Experiment 14.1.1

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Algorithm –

1. Start
2. Read two integers real1 and imag1 (first complex number)
3. Read two integers real2 and imag2 (second complex number)
4. Compute:
5. $real = real1 + real2$
6. $imag = imag1 + imag2$
7. If $imag \geq 0$
8. Print: $real + imag\ i$
9. Else
10. Print: $real - |imag|\ i$
11. End

Flowchart –



14.1.1 Complex Number Addition using Class

06:43 A L Z ? -

Write a Python program using a class named **Complex** to add two complex numbers.

Create a class **Complex** with the following methods:

- **initComplex()**: Read user input for the real and imaginary parts.
- **sum()**: Compute the sum of two complex numbers and store the result in the current instance.
- **display()**: Display the complex number in the format "<a> + i" or "<a> - i".

The program should create three instances of **Complex** class: two for input complex numbers and one for the result. Initialize both complex numbers, compute their sum, and display the result.

Input Format:

- First line contains two space-separated numbers: real and imaginary parts of first complex number
- Second line contains two space-separated numbers: real and imaginary parts of second complex number

Output Format:

- Print a single line showing the sum of the two complex numbers in the format "<a> + i" or "<a> - i".

Sample Test Cases

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Explorer

```
1 class Complex:
2     # Write the code.....
3     def initComplex(self):
4         self.real, self.imag = map(int, input().split())
5     def sum(self, c1, c2):
6         self.real = c1.real + c2.real
7         self.imag = c1.imag + c2.imag
8     def display(self):
9         if self.imag >= 0:
10            print(f"{self.real} + {self.imag}i")
11        else:
```

Average time
0.009 s
8.75 ms

Maximum time
0.014 s
14.00 ms

4 out of 4 shown test case(s) passed
4 out of 4 hidden test case(s) passed

Test case 1 14 ms Debug

Actual output

Expected output

3 4

2 5

5 + 9i

3 4

2 5

5 + 9i

Test case 2 9 ms

Terminal Test cases

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